

Math 150B Exam 1

September 18, 2025

Name: _____

I have read and understand the directions below, and I pledge that I have neither given nor received any unauthorized assistance on this exam.

(signature)

Directions:

1. Show all your work on this test paper and clearly indicate your answers.
2. If you have to make an argument, explain it using complete sentences. If we don't see the logical progression of your work, then we can't give you partial credit if the final answer is incorrect.
3. No scratch paper from the outside is allowed. Scratch paper will be provided by the instructor. Turn in all scratch paper to the instructor when you turn in your test.
4. Put all your final work on the test itself. Your scratch paper will not be graded.
5. Check that your test has 6 pages and 5 problems.
6. Make sure you sign the pledge.

Question:	1	2	3	4	5	Total
Points:	10	10	10	10	10	50
Score:						

1. (a) (3 points) $\int \frac{x}{1+x^2} dx$

(b) (3 points) $\int x \sin(x^2) dx$

(c) (4 points) $\int x\sqrt{x-2} dx$

2. (a) (5 points) Compute the area bounded by the curves $y = e^{3x}$, $y = e^{4x}$, $x = 0$ and $x = 1$.

- (b) (5 points) Compute the area bounded by the curves $x + y^2 = 1$ and $x + y = 0$.

3. (a) (5 points) Compute the volume of the solid where the base is the region enclosed by $y = x^2$ and $y = 9$. Slices perpendicular to the x -axis and parallel to the y -axis are squares.

- (b) (5 points) Compute the volume of the solid obtained by rotating the region bounded by $y = \sqrt{x}$ and $y = x^2$ about the line $x = 2$.

4. (a) (5 points) Compute the arc-length of the curve $y = x^{\frac{3}{2}}$ from $(0, 0)$ to $(1, 1)$.
- (b) (5 points) Compute the surface area of the surface generated by revolving about the y -axis the curve $y = x^2$ from $x = 0$ to $x = 2$.

5. (a) (2 points) Find the mass of a wire that is 2 ft long (starting at $x = 0$) and has a density function of $\rho(x) = x^2 + 2x$ lb/ft.
- (b) (4 points) A 1-m spring requires 10 J to stretch the spring to 1.1 m. How much work would it take to stretch the spring from 1 m to 1.2 m?
- (c) (4 points) Assume a tank in the shape of an inverted cone, with height 12 ft and base radius 4 ft. The tank is full to start with, and water is pumped over the upper edge of the tank until the height of the water remaining in the tank is 5ft. Set up the integral that would compute the work required to pump out that amount of water? Use 62.4 lb/ft^3 as the weight-density of water.

cheat sheet

Substitution rule: If $u = g(x)$, then

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

Basic trig derivatives

$$(\sin(x))' = \cos(x)$$

$$(\cos(x))' = -\sin(x)$$

$$(\tan(x))' = \sec^2(x)$$

$$(\sec(x))' = \sec(x)\tan(x)$$

Basic inverse trig derivatives

$$(\arctan(x))' = \frac{1}{1+x^2}$$

$$(\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}$$

$$(\operatorname{arcsec}(x))' = \frac{1}{|x|\sqrt{x^2-1}}$$

Basic algebra:

$$r \ln(x) = \ln(x^r)$$

$$\ln(AB) = \ln(A) + \ln(B)$$

$$e^{x+y} = e^x e^y$$

$$x^2 + 2bx + c = (x+b)^2 + c - b^2$$

Hooke's law: $F = kx$, where x is the displacement from the natural length and k is the spring constant.